Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

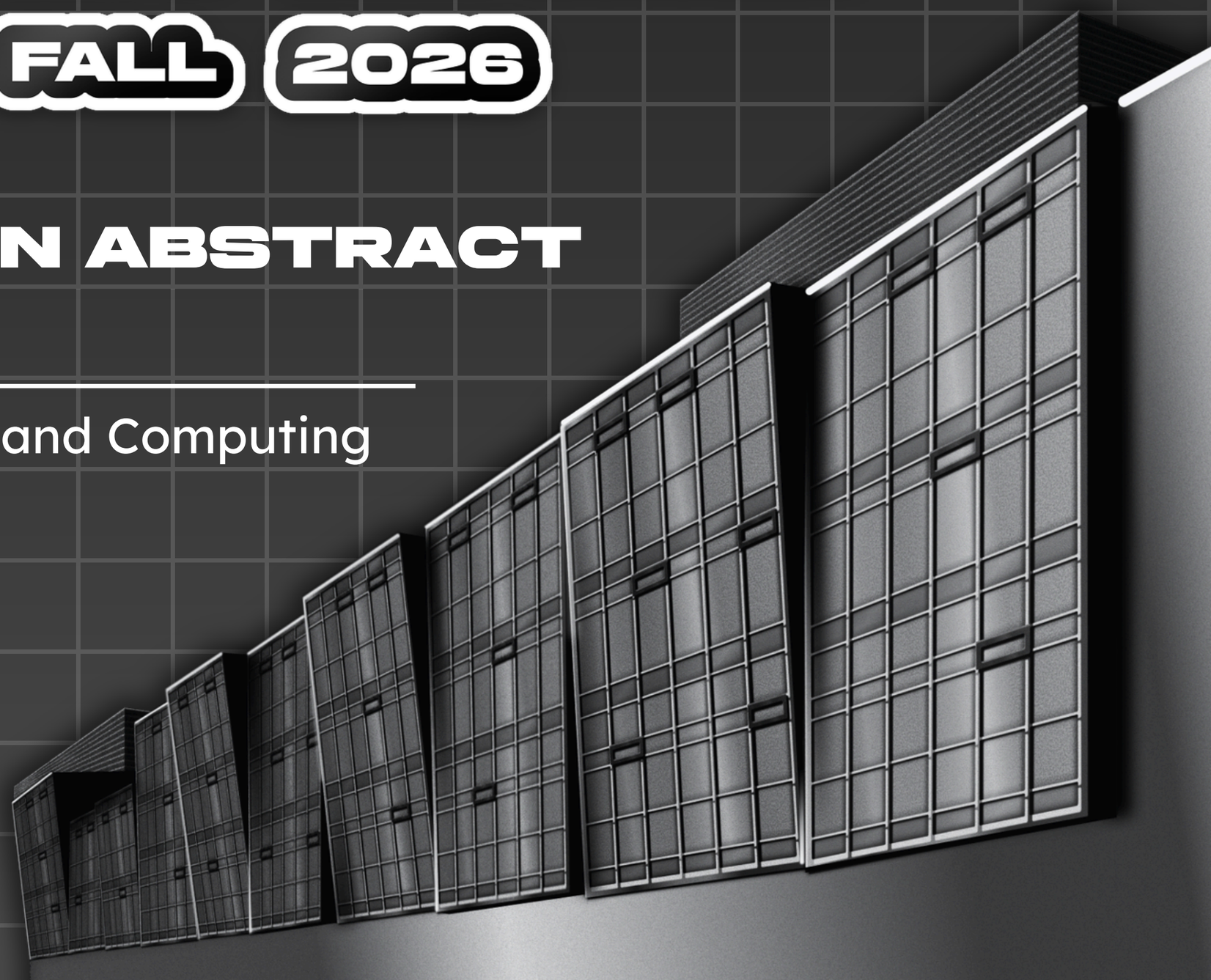
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

SENIOR DESIGN ABSTRACT

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



Metropolitan Engineering Competition 2026
Senior Design Competition Package
Smart Access Controlled Micro Vending Machine
Feeding the Future: Designing Smart Infrastructure for Urban Food Security

1.INTRODUCTION

Welcome to the Metropolitan Engineering Competition (MEC) 2026! MEC provides engineering students with the opportunity to apply their technical knowledge, creativity, and problem-solving skills to practical engineering challenges.

The Senior Design Competition challenges participants to integrate mechanical, electrical, and software engineering principles into functional systems through hands-on design, construction, programming, and testing.

This year's theme, “Feeding the Future: Designing Smart Infrastructure for Urban Food Security,” emphasizes the importance of accessible, reliable, and sustainable food infrastructure.

Through this competition, participants will explore how automated food-distribution systems can improve convenience, accessibility, and operational reliability while addressing practical constraints involving materials, power, and system integration.

2. THE CHALLENGE

2.1 Motivation

Automated food-distribution systems play an important role in providing convenient access to packaged food in locations such as universities, hospitals, workplaces, and community facilities.

However, developing a reliable automated dispensing system requires more than simply storing products inside an enclosure. Engineers must consider mechanical reliability, product handling, user interaction, electrical power, environmental conditions, and overall system safety.

Power interruptions introduce additional challenges, particularly when a system must continue providing essential services without its primary electrical supply.

The Senior Design Competition challenges participants to address these engineering problems by developing a compact automated food-dispensing system that integrates mechanical construction, electronic control, environmental monitoring, and backup-power functionality.

2.2 Description

Teams are required to design, construct, program, and demonstrate a functional tabletop micro vending machine capable of storing and dispensing standardized packaged snack products.

The machine must accommodate at least three (3) full-size packaged snack products representing three (3) different product selections.

Competitors must develop a system that allows users to select a product and receive the intended item through an automated dispensing mechanism.

The completed machine must integrate a programmable controller, user input, feedback indicators, environmental monitoring, and primary and backup power systems.

Teams may develop their own mechanical dispensing solutions using approved materials and components. No specific dispensing mechanism is required.

Objective	Description
Product Storage	Safely accommodate at least three full-size packaged snack products.
Product Selection	Provide three distinct product selections.
Dispensing Reliability	Release the correct product without unintended additional dispensing.
User Interaction	Provide an understandable and reliable selection interface.
Electrical Integration	Integrate the controller, sensors, actuators, and power systems.
Environmental Monitoring	Measure and communicate internal temperature and humidity.
Backup Operation	Maintain essential system functionality when primary power is unavailable.
Engineering Design	Demonstrate effective mechanical, electrical, and software integration.
Material Efficiency	Use available materials and the Engineering Dollar budget effectively.

Sustainability	Consider practical resource use, maintainability, and potential real-world applications.

Teams will have ten (10) hours to develop their solutions, construct and test their prototypes, and prepare their required deliverables.

During official testing, judges will evaluate the vending machine's functionality, dispensing accuracy, user interaction, environmental monitoring, power management, and engineering quality.

2.3. Requirements and Constraints

Requirement	Description
Team Size	Three (3) to four (4) competitors
Working Time	Ten (10) hours
Maximum Dimensions	30 cm × 25 cm × 45 cm, including permanently attached components
Starting Budget	200 Engineering Dollars (MEC$)
Product Capacity	At least three (3) full-size packaged snack products
Product Variety	Three (3) distinct selectable products
Controller	Arduino UNO R3 or ESP32
Dispensing	One intended product per valid selection
User Input	Keypad, push buttons, RFID, or approved combination
Feedback	Visible or audible system-status indication
Environmental Monitoring	DHT22 temperature and humidity sensor
Primary Power	Organizer-approved 12 V supply
Backup Power	Organizer-approved 9 V battery system
Power Switching	Automatic or manual switching permitted

Cooling	Optional fan-assisted ventilation or passive environmental management
Materials	Only organizer-approved components and construction materials
Safety	Secure wiring, stable structure, and safe moving mechanisms
Commercial Mechanisms	Ready-made vending mechanisms are prohibited

The Competition Organizers will designate three standardized packaged snack products for official testing. Every team will be evaluated using the same product types and package sizes.

The exact products, including their package dimensions and approximate masses, will be confirmed by the Competition Organizers and communicated equally to all teams before the design period begins.

Teams must design their machines to accommodate at least three (3) full-size packaged snack products representing three (3) different product types. Each product type must have its own selectable option. Teams may use separate dispensing mechanisms or a shared mechanism, provided the correct product is released for each selection.

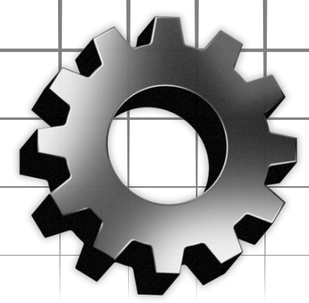
Automatic power switching is encouraged and will receive higher marks than manual switching when the required backup functionality is successfully demonstrated.

Teams may incorporate optional enhancements such as wireless connectivity, inventory monitoring, diagnostic displays, or advanced user interfaces. These features may strengthen the evaluation of the prototype within the existing judging criteria but will not receive additional points beyond the published 100-point rubric. Optional features must not replace mandatory functionality.

Design Flexibility

Teams may use gravity-fed mechanisms, servo-controlled gates, rotating components, sliding mechanisms, or other original designs to dispense products. No particular dispensing mechanism is required.

Active refrigeration is not required. Teams may incorporate fans, insulation, or other approved passive environmental-management features. The DHT22 sensor must be used to monitor temperature and humidity within the enclosure.



The primary objective is to produce a reliable, functional vending machine. Additional features will be considered only when the required systems operate successfully.

2.4. Design Restrictions and Safety

The controller must be an Arduino UNO or an ESP32. No other controllers are permitted.

Mechanical dispensing freedom: teams may use servo flaps, gravity gates, rotating wheels, push arms, carousels, chutes, sliding mechanisms, or another original approach. A simple, reliable gate can score strongly if it solves single-item release well. Commercial ready-made vending mechanisms are not permitted.

Active refrigeration, Peltier modules, and phase-change materials (PCM) are not permitted.

The vending machine must use the organizer-approved 12 V primary power supply and the designated 9 V backup battery system. Teams may implement automatic power switching using an approved switching module or manual switching using an approved toggle switch.

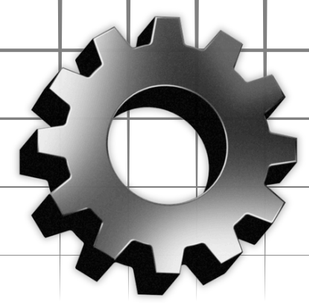
Teams must use appropriate voltage regulation and electrical connections to protect their components. Automatic switching is encouraged and will receive higher marks than manual switching when both systems meet the required functionality.

The vending machine must demonstrate essential electronic functionality using the designated 9 V backup battery when the primary 12 V power supply is unavailable.

During backup operation, the microcontroller must remain functional, the DHT22 temperature and humidity monitoring system must remain operational, and the machine must provide a clear visual or audible indication that backup power is active. Any LCD display or LED indicators used to provide this backup-status indication must remain functional. User selection through a keypad, RFID system, or other input interface is not required during backup operation. Snack dispensing using backup power is optional.

Successful dispensing of at least one designated snack using backup power is encouraged and will be recognized as an advanced functional achievement within the existing judging criteria. Backup dispensing is not mandatory.

Both manual and automatic power-source switching are permitted. Automatic switching will receive higher marks when implemented successfully and safely. Optional cooling fans and other nonessential accessories are not required to operate using backup power.



The completed prototype must not exceed 30 cm × 25 cm × 45 cm in its normal operating configuration, including all permanently attached components. External power adapters and detachable programming cables supplied for testing are excluded from the dimensional measurement. The prototype must satisfy these requirements during official inspection and testing.

Wiring must be secured and moving mechanisms must be safe. Teams are expected to maintain safe electrical practices when wiring and operating their prototypes. The system must complete all demonstration steps within the designated time limit during evaluation.

3. RULES

3.1. Allowed Resources

Competitors may use textbooks, course notes, programming documentation, and other references permitted by the Competition Organizers.

Each competitor may use one (1) electronic device at a time.

Internet access is permitted for research, programming documentation, and approved technical assistance.

Only approved competition materials may be incorporated into the completed prototype.

Competitors may not bring external construction materials, preassembled mechanisms, or completed prototype sections unless expressly authorized by the Competition Organizers.

3.2. Question Periods

Following the competition briefing, competitors may ask questions regarding the challenge, technical requirements, materials, testing procedures, and judging criteria.

During the competition, questions must be directed to the Competition Leads or submitted through the official clarification process.

Clarifications that materially affect competition requirements will be communicated consistently to all teams.

3.3. Testing Rules

Each team will have ten (10) minutes to complete its official prototype testing. This period includes product selection and dispensing tests, environmental monitoring, and primary-to-backup power demonstrations. Teams must ensure that their prototypes are ready for testing when their assigned evaluation period begins.

All teams will be evaluated using standardized testing conditions and procedures.

Before official testing, each vending machine must undergo an inspection conducted by the Competition Organizers.

The inspection will verify the prototype's dimensions, approved materials, electrical safety, structural stability, and compatibility with the designated testing equipment.

All permanently attached components must be included when measuring the prototype's external dimensions.

Teams whose prototypes fail to satisfy mandatory safety or dimensional requirements must correct the identified issues within the permitted competition period before participating in official testing.

Product Storage and Selection Test

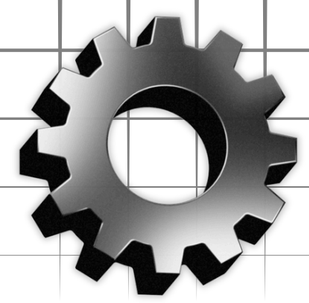
- Each machine must accommodate at least three (3) standardized full-size packaged snack products representing three (3) different product selections.
- Teams must demonstrate that each product can be selected using the machine's intended user interface.

Dispensing Test

- Each team will be required to demonstrate one (1) official dispensing attempt for each of the three (3) designated product selections. Each attempt will begin with the product loaded and the machine in its normal operating state. Judges will record whether the correct product is dispensed successfully. Additional attempts may be permitted for demonstration purposes at the discretion of the Competition Organizers but will not replace the recorded official results.
- A successful dispensing operation occurs when the user selects one of the three designated products and the machine releases exactly one corresponding product into an accessible retrieval area without requiring the user or organizers to manually operate the dispensing mechanism. Unintended additional releases, incorrect products, or products that remain trapped inside the mechanism will not count as successful dispensing operations.
- Judges will evaluate dispensing accuracy, reliability, product handling, and the need for manual intervention.

Environmental Monitoring Test

- Teams must demonstrate functional temperature and humidity monitoring using the DHT22 sensor.
- Sensor readings must be observable or recorded during the demonstration.



- Optional ventilation, insulation, or environmental-management features may be demonstrated and explained.

Primary and Backup Power Test

- Teams must demonstrate normal operation using the organizer-approved primary power supply.
- Under organizer supervision, the primary power source will be disconnected or disabled to evaluate backup operation.
- Teams may use automatic or manual power-source switching.
- During backup-power testing, the Competition Organizers will disconnect or disable the primary 12 V power supply. Judges will verify that the vending machine operates using the designated 9 V backup battery.
- The team must demonstrate that the microcontroller remains functional, the DHT22 continues providing temperature and humidity readings, and the machine provides a clear visual or audible indication that backup power is active. Any LCD display or LED indicators used for this backup-status indication must remain operational. User selection and snack dispensing are not required during backup-power testing.
- Teams may use manual or automatic power-source switching. Automatic switching will receive higher marks when demonstrated successfully.
- Teams that can also dispense at least one designated product using backup power will receive additional recognition within the existing Functional Testing criteria. Following the backup demonstration, the primary power supply will be restored and judges will verify that the machine returns to normal operation.
- The cooling fan and other nonessential accessories are not required to operate on backup power.
- Automatic switching will receive higher marks than manual switching when the required functionality is demonstrated.

Official testing will be conducted according to the schedule established by the Competition Organizers.

All teams will receive equivalent testing opportunities and will be evaluated using the same judging criteria.

3.4. Judging Overview

Category	Maximum Points
Design Quality	30
Functional Testing	40
Presentation	30
TOTAL	100

3.5. Competition Penalties

Violation	Points Deducted
Plagiarism	–50
Insufficient citations	–50
Documents received after the official deadline	–50
Absent team member during a required competition activity	–25

Penalties will be applied according to the official Senior Design judging rubric and MEC competition regulations.

4. DELIVERABLES

Each team must complete the following deliverables.

Physical Prototype

A completed tabletop micro vending machine satisfying the competition's dimensional, functional, electrical, material, and safety requirements.

Functional Demonstration

Participation in the official testing procedure, including product selection, dispensing, environmental monitoring, and primary-to-backup power operation.

Engineering Presentation

A PDF presentation explaining the team's design process, mechanical construction, electrical architecture, programming approach, testing observations, engineering decisions, and proposed improvements.

Code Submission

Teams must submit the source code used to operate their prototype. Arduino programs should be submitted as .ino files, while other approved source files may be submitted in their original programming format or as .txt files.

Files must follow the naming convention:

MEC2026_TeamName_SeniorDesign_Code.ino

or

MEC2026_TeamName_SeniorDesign_Code.txt

Teams must ensure that the submitted files contain the complete code required to demonstrate their prototype.

4.1. Final Presentation

Each team will have twenty-five (25) minutes for evaluation, consisting of a ten (10) minute engineering presentation, a ten (10) minute prototype testing period, and a five (5) minute question-and-answer period.

All team members must attend and participate.

The presentation must explain the team's design process, engineering decisions, technical integration, prototype functionality, and testing observations.

Presentations must be submitted as PDF files.

The official submission method, deadline, and presentation schedule will be communicated by the Competition Organizers.

4.2. Plagiarism and Academic Integrity

All submitted designs, prototypes, code, and presentations must represent the team's own work completed during the competition period.

External references, images, technical resources, and design inspiration must be appropriately acknowledged.

AI tools may be used for permitted assistance, but teams remain responsible for understanding, verifying, and explaining their submitted work.

Plagiarism, unauthorized pre-developed solutions, or deliberate misrepresentation may result in penalties under official MEC regulations.

4.3 Final Testing

Official functional testing will take place during the ten (10) minute testing portion of each team's twenty-five (25) minute evaluation period. Teams must complete the required demonstrations outlined in Section 3.3 within this time. Testing will be supervised by the Competition Organizers and judges.

5. MATERIALS

The Metropolitan Engineering Competition will provide each Senior Design team with an initial set of materials and access to the MEC Shop. Each team will receive 200 Engineering Dollars (MEC$) to purchase additional approved components and construction materials. Available items, prices, and purchase limits are outlined in the following sections.

5.1 Materials Provided

The following materials and tools will be provided to each Senior Design team at no Engineering Dollar cost.

Materials	Quantity Given per Team	Return or Keep
Mini Hot Glue Gun	1	Return
Scissors	1	Return

X-Acto / Utility Knife	1	Return
Plastic Ruler	1	Return
9 V Battery	1	Keep
Mini Hot Glue Sticks	10	Keep
HB #2 Pencils	1	Keep
SL2.5 Mini Flat-Head Screwdriver	1	Return
Electrical Tape	1	Keep
Half-Sized Breadboard	1	Keep
Eraser	1	Keep

5.2 MEC Shop Items

Each Senior Design team will receive **200 Engineering Dollars (MEC$)** to purchase approved materials and electronic components from the MEC Item Shop.

All available items have assigned Engineering Dollar prices and may be subject to purchase limits and inventory restrictions.

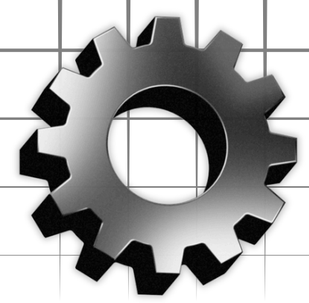
Teams must remain within their allocated budget. Unused Engineering Dollars will not be exchanged for additional competition points or other benefits.

Materials are available in limited quantities. Teams are encouraged to plan their purchases carefully and consider alternative components when necessary.

Ordering procedures, purchase restrictions, and applicable refund or exchange policies will be communicated by the Competition Organizers.

Certain pre-given materials, including mini hot glue sticks, may be available for additional purchase through the MEC Shop. Additional purchases are subject to the listed Engineering Dollar prices, purchase limits, and remaining inventory.

Servo motors are available in different rotation configurations. Teams are responsible for selecting components appropriate for their dispensing mechanisms.



Each team may purchase a maximum of one (1) microcontroller in total, either an Arduino UNO R3 or an ESP32.

Each team will receive one half-sized breadboard at no cost. Additional full-sized breadboards may be purchased through the MEC Shop.

Each team will receive one (1) 9 V battery for the required backup-power demonstration. Additional batteries will not be automatically provided and may only be obtained through an organizer-approved process, subject to availability.

6. APPENDIX

6.1. MEC Shop Price List

The following materials are available for purchase through the MEC Item Shop. All prices are listed in Engineering Dollars (MEC$). Items are subject to availability and the maximum purchase limits indicated below.

Item	Price (MEC$)	Maximum Per Team
Microcontroller - ESP32 or Arduino UNO R3	20	1 total
LED Matrix Display Module	15	1
RFID Reader Module with Tag/Card	10	1
DHT22 Temperature & Humidity Sensor	7	1
USB Cable - 0.5 m	3	2
USB Type-C to Type-B Arduino Cable	4	1
LM2596 Step-Down Voltage Regulator	8	1
9 V Battery Connector	2	2
SG90 Micro Servo Motor - 180°	7	2
SG90 Micro Servo Motor - 360°	7	2
3×4 Membrane Keypad	8	1

12 mm Push Button	3	3
TCS3200 Colour Sensor	10	1
Breadboard Power Supply Module	6	1
830-Point Breadboard	8	1
Breadboard Jumper Wires - 20 Piece Pack	4	2
Dupont Jumper Wires - 10 Piece Pack	4	3
1 kΩ Resistors - 3 Piece Pack	3	4
LEDs - 3 Piece Pack	3	3
16×2 LCD Display with I2C Backpack	12	1
120 mm 12 V Cooling Fan	12	1
Active Buzzer - 5 V or 3 V	4	1
Screw Terminal Block	2	5
Clear Plastic Sheet	7	1
12 V 2 A Power Supply Adapter	14	1
MTS-102 SPDT Toggle Switch	4	2
Dual DC Automatic Power Switching Module	15	1
White Foam Board - 20 × 30 in.	8	3
Black Foam Board - 20 × 30 in.	10	1
Double Reflective Insulation - 1 ft	8	2
Coloured Paper Straws - 2 Piece Pack	3	2
Corrugated Cardboard Sheet	6	4
Mini Hot Glue Sticks - 2 Piece Pack	6	2

Thin Twine String - 3 ft	4	3
Small Fridge Magnet	3	5
Silicon Carbide Sandpaper	4	3
Mini Metal Hinges - 2 Set Pack	4	3

Purchase limits apply to each team unless otherwise specified. The combined purchase limit for Arduino UNO R3 and ESP32 microcontrollers is one (1) controller per team. For servo motors, the maximum of two (2) applies separately to each listed rotation type. All purchases are subject to available inventory.

6.2. Standardized Testing Specifications

The following specifications establish the requirements for official Senior Design testing. All teams will be evaluated under equivalent conditions using organizer-approved equipment and procedures.

Component	Specification
Prototype	Tabletop micro vending machine
Maximum Dimensions	30 cm × 25 cm × 45 cm
Product Capacity	Minimum three full-size packaged snacks
Product Selections	Three distinct selections
Standardized Products	Organizer-designated packaged snack products
Controller	Arduino UNO R3 or ESP32
Dispensing	One intended product per valid command
Environmental Monitoring	DHT22 temperature and humidity sensor
Primary Power	Organizer-approved 12 V supply
Backup Power	Organizer-approved 9 V battery
Power Switching	Automatic or manual

Backup Requirement	Functional controller, DHT22 monitoring, and visual or audible backup-status indication. User selection and snack dispensing are optional during backup operation.
Cooling Fan on Backup	Not required
Testing Conditions	Equivalent procedures for all teams
Official Testing Schedule	Communicated by Competition Organizers